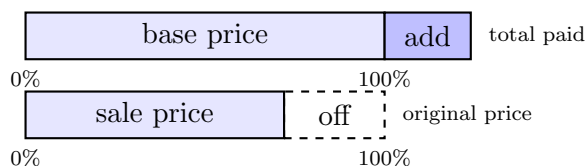


Percent Applications: Tax, Tip, Discount, and Markup

Grades 6–7 Ratios and Proportional Relationships

Sales tax, tips, discounts, markup, and working backwards to the original price

The percent bar model. Every problem on this sheet is one of these two pictures. The whole bar is the *base* — the price the percent is taken *of*. Adding tax, a tip, or a markup builds a bar longer than the base; a discount cuts the bar shorter.



A model of the two situations. No values shown: use the numbers given in each problem.

Directions.

- Name the base first — the price the percent is taken of — then decide whether the percent is added on or taken off.
- Show the multiplier you use (for example 1.08 or 0.85) or the two steps you take.
- All prices are in dollars. Round money answers to the nearest cent and write the answer on the answer line.

1. A jacket is priced at \$40. The sales tax rate is 6%. How much sales tax is charged, in dollars? (Find the tax only, not the total.)

Answer: _____

2. A family's dinner bill is \$36 before the tip. They want to leave a 15% tip. How much is the tip, in dollars?

Answer: _____

3. A backpack is priced at \$45 and is on sale for 20% off. Fill in the ratio table, then write the sale price in dollars.

percent of price	100%	
dollars		

Answer: _____

4. A bike shop buys a bike from the maker for \$80 and marks the price up by 35% before putting it on the floor. What is the shop's selling price, in dollars?

Answer: _____

5. A calculator is priced at \$24.50 and the sales tax rate is 7.25%. What is the total amount paid, in dollars, rounded to the nearest cent?

Answer: _____

6. A restaurant bill is \$58.40 before the tip. A customer leaves an 18% tip. What is the total amount the customer pays, in dollars, rounded to the nearest cent?

Answer: _____

7. A coat is priced at \$72. It is marked 25% off, and then 6% sales tax is charged on the sale price. What is the total amount paid, in dollars? Take the two steps in order — the tax is charged on the sale price, not on the original price.

Answer: _____

8. A store pays \$18.60 for a board game and marks it up 45%. What is the shelf price, in dollars, rounded to the nearest cent?

Answer: _____

9. A sweater is on sale for \$34 after a 15% discount. Let x be the original price in dollars. Write an equation for x using the sale price, then solve it to find the original price in dollars.

Answer: _____

10. A receipt shows a total of \$77.04, which already includes 7% sales tax. Let x be the price before tax, in dollars. Write an equation for x , then solve it to find the price before tax in dollars.

Answer: _____